Solution to XOR-bitant!

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1. *Show that there is a unique number e base m such that*

$$a\overline{\oplus}^m e = e\overline{\oplus}^m a = a.$$

First, we see that $e = 0$ clearly satisfies this property. To show that this value is unique, suppose that some nonzero number f also satisfies this property. Then, there is some digit k whose value is between 1 and $m-1$. Let the corresponding digit of a be ℓ; then, $k \oplus \ell \not\equiv \ell$, which means the result of $a\overline{\oplus}^m f$ differs from a in at least one digit, so there is no such f.

2. *Is there always a number a' such that $a\overline{\oplus}^m a' = e$?*

Given some number a in base m, we can determine a value of a' satisfying the above equation as follows: The nth digit of a' is the additive inverse modulo m of the nth digit of a. For example, if $a = 239871_{10}$, then the value of a' satisfying

$$a\overline{\oplus}^{10} a' = e$$

would be 871239_{10}. This value of a' has the property mentioned in the problem because the corresponding digits must combine to 0 after applying $\oplus^m$, so a' exists for every a.

3. *Does $\overline{\otimes}^m$ satisfy the same properties as $\overline{\oplus}^m$? If not, are there any restrictions we can impose on a, b, and/or m for which these properties are satisfied?*

No, this operation doesn't have the two properties mentioned above. This is because there can be several values of e for which

$$a\overline{\otimes}^m e = a.$$

For example, let $a = 1234_7$. Then,

$$1234_7\overline{\otimes}^7 1111_7 = 1234_7\overline{\otimes}^7 11111_7 = 1234,$$

so both 1111_7 and 11111_7 have this property. In fact, any string of 1s with at least as many digits as a will return a upon combination by $\overline{\otimes}^m$, since the excess leading 1s will just get cancelled out by the leading zeros in a.

If we take e to be the string of 1s with the same number of digits as a, however, then there will be an a' for which $a\overline{\oplus}^m a' = e$ when $gcd(f_n(a), m) = 1$, where $f_n(k)$ is the nth digit of k and n is any positive integer less than or equal to the number of digits in a. This is because, if $a\overline{\oplus}^m a' = e$, then $f_n(a) \cdot f_n(a') \pmod{m} = 1$, which means $f_n(a')$ is the modular inverse of $f_n(a)$, which exists if and only if $gcd(f_n(a), m) = 1$.

4. *Do $\overline{\oplus}^m$ and $\overline{\otimes}^m$ satisfy the distributive property? That is, does*

$$a\overline{\otimes}^m (b\overline{\oplus}^m c) = (a\overline{\otimes}^m b)\overline{\oplus}^m (a\overline{\otimes}^m c)?$$

The answer is yes. Let $f_n(k)$ be the nth digit of k. Then,

$$\begin{aligned}
f_n(a\overline{\otimes}^m(b\overline{\oplus}^m c)) &= f_n(a)\cdot f_n(b\overline{\oplus}^m c) \pmod m\\
&= f_n(a)\cdot(f_n(b)+f_n(c)) \pmod m\\
&= (f_n(a)\cdot f_n(b))+(f_n(a)\cdot f_n(c)) \pmod m\\
&= f_n((a\overline{\otimes}^m b)\overline{\oplus}^m(a\overline{\otimes}^m c)),
\end{aligned}$$

which means the nth digit of $a\overline{\otimes}^m(b\overline{\oplus}^m c)$ is equivalent to the nth digit of $(a\overline{\otimes}^m b)\overline{\oplus}^m(a\overline{\otimes}^m c)$ regardless of the value of n. Hence, the two numbers are equivalent, which implies that $\overline{\otimes}^m$ is distributive over $\overline{\oplus}^m$.